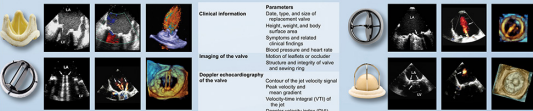




Normal Flow Patterns and Essential Parameters in the Evaluation of Prosthetic Valves



Prosthetic Aortic Valves

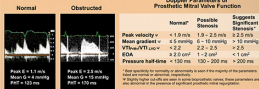


Comparison with previous
post-operative study,
when available

Contour of the jet velocity signal
Peak velocity and
mean gradient
Velocity-time integral (VTI) of
the jet
Doppler velocity index (DVI)
Pressure half-time (PHT) in
mitral and tricuspid valves
Effective orifice area (EOA)
Presence, location, and severity
of regurgitation
Comparison of above parameters
helpful in suspected valvular
dysfunction

Prosthetic Mitral Valves

Doppler Parameters of Prosthetic Mitral Valve Function

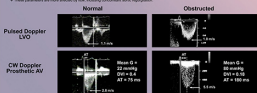


Doppler Parameters of Prosthetic Aortic Valve Function in Mechanical and Stented Biological Valves*

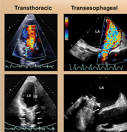
	Normal	Possible Stenosis	Suggests Significant Stenosis
Peak velocity v	< 3 m/s	3.4 m/s	> 4 m/s
Mean gradient $\bar{v}$	< 20 mmHg	20–35 mmHg	> 35 mmHg
Doppler velocity index	> 0.30	0.29–0.25	< 0.25
Effective orifice area	> 1.2 cm ²	1.2–0.8 cm ²	< 0.8 cm ²
Contour of the jet velocity	Triangular, Early peaking	Triangular to Intermediate	Rounded, symmetrical contour
Acceleration time	< 80 ms	80–120 ms	> 100 ms

* In conditions of normal or near normal stroke volume (30–70 ml).

* These parameters are more affected by flow, including concomitant aortic regurgitation.



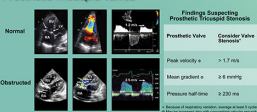
Flow masking in mechanical valves from the transaortic approach
can hinder assessment of prosthetic mitral regurgitation



Findings Suggestive of Significant Prosthetic Mitral Regurgitation by TTE in Mechanical Valves with Normal Pressure Half-time

- Peak mitral velocity ≥ 1.9 m/s
- VTI_{mitr}/VTI_{aor} ≥ 2.5
- Mean gradient ≥ 5 mmHg
- LV stroke volume derived by 2D or 3D is $\geq 30\%$ higher than systemic stroke volume by Doppler
- Systolic flow convergence seen in the LV towards the prosthesis
- Tricuspid regurgitation jet velocity > 3 m/s

Prosthetic Tricuspid Valves



Prosthetic Pulmonic Valves

